UCLA Samueli School of Engineering

Budhaditya (Bradlee) De, E.I.T

Department of Civil and Environmental Engineering, UCLA, Los Angeles, CA-90085

budhaditya@g.ucla.edu +1 (310) 307-9195 LinkedIn Github

PROFESSIONAL SUMMARY

- 4th Year Ph.D. candidate with a passion for Structural mechanics, Computational modeling, Analysis and seismic-resistant design and retrofit of various structural systems including Reinforced Concrete, Steel & Timber.
- Particular interests in Nonlinear structural response, Probabilistic structural risk, reliability and resilience, Performance-based hazard engineering, Vibration control and AI/ML applications in structural mechanics.
- Proficiency in High-fidelity FE simulations, Computer programming, Uncertainty quantification and Optimization. Soft skills include collaborative and teamwork mindset, quick rapport building and clear communication.

EDUCATION

Ph.D in Civil & Environmental Engineering - University of California, Los Angeles [GPA: 4.0/4.0] ***Spring 2027***

Major: Structural Earthquake Engineering *Minor:* Mechanical Engineering and Statistics

M.S in Civil Engineering - Indian Institute of Technology Kanpur [GPA: 10.0/10.0] ***June 2023***

Major: Structural Engineering and Computational Mechanics

B.S in Civil engineering - Jadavpur University [GPA: 9.42/10.0] ***June 2021***

Specialization: Structural Engineering and Mechanics

WORK/RESEARCH EXPERIENCE

Graduate Student Researcher (GSR)/ Ph.D. Candidate ***August 2023-present***

University of California, Los Angeles (UCLA), Los Angeles, CA

- Proposed a probabilistic framework for nonlinear structural model updating using visual damage and measured residual drifts for realistic post-earthquake damage evaluation.
- Developing an AI-augmented framework for probabilistic post-earthquake structural demand, damage and risk evaluation on regional scale through Bayesian networks using multiple data modalities.

Assistant Structural Designer (Intern) ***June 2026-September 2026***

Degenkolb Engineers, Los Angeles, CA

- Seismic evaluation and retrofit of healthcare and academic buildings per ASCE 41-23 using nonlinear analysis.
- Developed python-based automated P-M_x-M_y interaction codes for RC walls and columns and demand-capacity ratio (DCR) visualization tool for RC beams using Visual Basic and MS Excel.
- Conducted Nonlinear pushover and dynamic analysis using soil springs on ETABS and Perform-3D along with foundation evaluations on CSI SAFE.

Graduate Research Assistant (GRA) ***August 2021-June 2023***

Indian Institute of Technology Kanpur (IITK), Kanpur, India

- Proposed zone-based design Pressure coefficients, Power Spectral Density and Coherence models for conical membranes using high-fidelity turbulent wind-structure interaction in Ansys Fluent using Large Eddy Simulation.
- Developed gust-response and nonlinear adjustment factors through nonlinear dynamic analysis in ABAQUS using stochastic wind loads for aiding the practical design of such structures.

Structural Engineering Intern (Summer research fellow) ***May 2020-July 2020***

Indian Institute of Technology Hyderabad (IITH), Hyderabad, India

- Studied properties of Fiber Reinforced Concrete (FRC) and influence of fiber orientations on its ductile behavior.

Structural Engineering Intern ***May 2019-June 2019***

Indian Institute of Technology Kharagpur (IITKGP), Kharagpur, India

- Analyzed vibration characteristics and frequency domain properties of steel portal frames, fiber-reinforced polymer composite plates and steel cantilevers through impact hammer testing & accelerometer monitoring.

RELEVANT ACADEMIC PROJECTS

Nonlinear analysis and performance-based design of a 21-story RC core wall building (*Link*) ***2024***

Course: C&EE 243B: Behavior and design of RC structural systems

- Modeled the structure in Perform-3D and conducted nonlinear response history analysis with a suite of 7 ground motions.

- Analyzed demand parameters (drifts, rotations, stresses and strains) within allowable limits per LATBSDC and ASCE 41 guidelines and designed the RC core wall using performance-based seismic design principles.

Seismic evaluation and retrofit of an existing 5-story RC building using ASCE 41-23 (*Link*) ***2024***
Course: C&EE 243B: Behavior and design of RC structural systems

- Conducted nonlinear pushover analysis in Perform-3D and evaluated various failure limit states using ASCE 41-23.
- Proposed an economic retrofit scheme (e.g. Concrete, FRP jacketing) within 30% of the total replacement cost.

Computer code for 2D/3D 1st order linear elastic structural analysis using the stiffness method (*Link*) ***2023***
Course: C&EE 235A: Advanced structural analysis

- Developed a stiffness-based computer code for 3D first order linear elastic structural analysis using MATLAB including various support conditions, loading profiles, temperature gradients, lack of fits etc.

Seismic-resistant design of a 5-story RC moment frame building using American standards ***2023***
Course: C&EE 234A: Behavior and Design of RC structural Elements

- Designed RC beams, slabs, columns, beam-column joints and diaphragms using ASCE 7-22, ACI 318-19

Design of a 5-story dual Shear Wall-Moment Frame office building using Indian standards (*Link*) ***2022***
Course: CEE 629A: Earthquake analysis and design of structures

- Modeled and analyzed the dual structural system in CSI SAP2000 under combined gravity and earthquake loads calculated using the ELF procedure.
- Designed earthquake-resistant beams, columns, beam-column joints and shear walls using design load combinations from the model with reference to IS 456:2000 and IS 1893:2016.

TECHNICAL SKILLS

- **Software:** CSI ETABS, SAP2000, Perform-3D, Bentley RAM, OpenSees, AutoCAD/Revit, MS 365 Suite, Bluebeam.
- **Programming/ Packages:** Python, MATLAB, Scikit-Learn, Keras, Tensorflow, PyTorch, C/C++.

SELECT HONORS & ACHIEVEMENTS

- **NSF CMMI Research Grant Scholarship** for the project "Multimodal Disaster Impact Assessment Models for Rapid Post-Hazard Decision Support and Enhanced Resilience. ***2023-present***
- **Degenkolb Engineers' Outstanding Student Merit Scholarship** of **$5000** USD for excellent academic performance and leadership activities in Graduate school. ***2026***
- **ASCE LA Section Young Member Forum (YMF) Outstanding Graduate Student Award and Scholarship** of **$2000** USD. ***2026***
- **3rd Place Finalist** in the Probabilistic Methods Committee (PMC) student paper competition, Engineering Mechanics Institute (EMI) Conference, Anaheim, California. ***2025***
- **ASR Sai Memorial Gold Medal** - 1st place by merit for Master in Structural Engineering degree, IIT Kanpur. ***2023***
- **Graduate Research Assistant** scholarship granted by Dean of Academic Affairs (DOAA), IIT Kanpur. ***2021-23***
- **Academic Excellence Award** - Outstanding Academic performance in Structural Engineering, IIT Kanpur. ***2021-22***
- **University Gold Medal** - 1st place by merit for Bachelor in Civil Engineering degree, Jadavpur University. ***2021***
- Secured **All India Rank 290** amongst approximately 115,000 applicants (Top 99.83 %) in the Graduate Aptitude Test in Engineering (GATE) for Civil Engineering in the first attempt. ***2021***
- Recipient of the **Summer Research Fellowship Program (SRFP)** fellowship, granted by the Indian Academy of Sciences (IAS). ***2020***
- **Indubhushan & Shantishudha Memorial Gold Medal** – 1st place by merit for undergrad freshman and sophomore years (AY 2018/19 & 2019/20) ***2019, 2020***
- **1st place Winner** for a popsicle suspension bridge building competition "**Bridge-The-Gap**" in the undergrad Technical Fest SRIJAN, Jadavpur University. ***2020***

RESEARCH PUBLICATIONS

- **De, B.**, Burton, H.V. (2026). "Multi-Fidelity Gaussian Process Surrogate for Probabilistic Regional Seismic Demand Estimation using Observed Residual Drifts." *Earthquake Engineering & Structural Dynamics* (*In Prep.*)
- **De, B**., Burton, H.V., Al Adsani, M. (2026). "Bayesian nonlinear structural model parameter updating using visual damage indicators." *Journal of Engineering Mechanics.* (*Link*)
- **De, B.,** Wu, C., Burton, H.V., Zhou, Z. (2026). "Residual Drift-informed Aftershock Damage and Risk Assessment

through Nonlinear Structural Model Parameter Updating." *Earthquake Engineering and Structural Dynamics.* (*Under Review*)

- **De, B.**, Dutta, D., Majumder, R., Mishra, S.K. (2026). "A Hybrid Gaussian Process-XG Boost Machine Learning Framework for Robust Knockdown Factor Prediction and Reliability-based Design of Thin Spherical Shell Caps." *Computers and Structures.* (*Under Review*)
- **De, B.**, Dutta, D., Majumder, R., Mishra, S.K. (2026). "Reliability-based Design Optimization of Thin Spherical and Truncated Conical Shells against Post-Critical Load Drops using Physics-informed Deep Learning Metamodels." *Computer-Aided Civil and Infrastructure Engineering.* (*In prep.*)
- Dutta, D., **De, B.**, Majumder, R., Mishra, S.K. (2026). "A Physics-informed Neural Network Approach for Robust Buckling Load Prediction and Reliability-Based Design of Thin Truncated Conical Shells." *arXiv preprint arXiv:2608.21818.* (***Link***)
- Majumder R., **De, B.**, Gupta, A.D/, Mishra, S.K. (2026). "A data-driven assessment of buckling strength reduction in truncated conical shells: Development of a hybrid Gaussian process-XG boost machine learning framework." *Data-Centric Engineering*; 7:e14. (***Link***)
- **De, B**., Burton, H.V. (2025). "Nonlinear structural model parameter updating using residual drift measurements from sequential seismic events." *Journal of Structural Engineering.* (***Link***)
- Kumar, A., **De, B.**, Mishra, S.K. (2026). "Wind-induced response and an equivalent static wind-resistant design for anticlastic conical tensile membranes." *Wind Engineering.* (***Link***)
- **De, B**., Kumar, A., & Mishra, S. K. (2024). "Large eddy simulation of wind loading on an anticlastic conical tensile membrane." *Journal of Wind Engineering and Industrial Aerodynamics*, *246*, 105658. (***Link***)
- **De, B**., Kumar, A., & Mishra, S. K. (2023). "A Comparative Study of Peak Factor Prediction Models for Aerodynamic Loading on an Anticlastic Conical Tensile Membrane Structure." *Recent Developments in Structural Engineering, Volume 3, 13th Structural Engineering Convention (SEC), VNIT Nagpur 2023.* (***Link***)

CONFERENCE PRESENTATIONS

- **13th Structural Engineering Convention (SEC), VNIT Nagpur, India**– "A Comparative Study of Peak Factor Prediction Models for Aerodynamic Loading on an Anticlastic Conical Tensile Membrane Structure." ***December 2023***
- **Engineering Mechanics Institute (EMI) Conference 2025, Anaheim, California**- "Bayesian Approach to Nonlinear Structural Model Parameter Updating using Visual Damage Indicators." ***May 2025***
- **14th International Conference on Structural Safety and Reliability (ICOSSAR), Los Angeles, California**- "Nonlinear structural model parameter updating using residual drift measurements from sequential seismic events." ***June' 25***
- **Engineering Mechanics Institute (EMI) Conference 2026, Boulder, Colorado**- "Residual Drift-informed Aftershock Damage and Risk Assessment through Nonlinear Structural Model Parameter Updating." ***June 2026***
- **NIST-NSF Disaster Resilience Research Symposium 2026, Rockville, Maryland**- "Multimodal Disaster Impact Assessment Models for Enhanced Resilience." ***August 2026***

RELEVANT COURSEWORK (*Please see Transcripts for more details*)

Finite Element Analysis, Nonlinear Structural Analysis, Structural Response to Earthquakes, Advanced Steel Structures Design, Advanced Concrete Design, Linear & Nonlinear Elasticity, Structural Stability, Energy and Variational Methods in Computational Mechanics, Analytical Fracture Mechanics, ML and AI applications in Civil Engineering, Monte Carlo Methods for Optimization, Stochastic Processes in Dynamical Systems.

EXTRA-CURRICULAR/ PROFESSIONAL ENGAGEMENT

- Programming lead (UCLA) and organizing committee member for the SoCal CEE research symposium ***2025***
- Student member of the American Society of Civil Engineers (ASCE). ***2024-present***
- Member of the EERI-SEAOSC student community from UCLA. ***2023-present***
- Teaching Assistant - Advanced Structural Analysis (UCLA), Structural Systems Design (UCLA), Design of Steel Structures (IIT Kanpur), Structural Stability (IIT Kanpur). ***2023-present***
- Host of 'Quizzard of Oz' and organizing committee member for 'Sanrachna', annual technical fest of the Dept. of Civil Engineering, Jadavpur University. ***2019***